Summary

- A transformation is a specific set of rules that change the preimage onto the image.
- A ***translation*** is a transformation that moves all points of a figure in the same distance in the same direction.
- In horizontal, the right-side movement is denoted by → and the left side movement is denoted by ←.
- In vertical, the upside movement is denoted by ↑ and the downward movement is denoted by ↓.
- A ***reflection*** is a transformation that "flips" or "reflects" a figure about a line.
- A ***rotation*** is a transformation that turns every point of the pre-image through a specified angle and direction about a point. The fixed point is called the centre of rotation. The angle is called the angle of rotation.
- A rotation is also called a turn.
- The default direction of a rotation is the anti-clockwise direction.
- Rotation of 360° is called a full turn, rotation of 180° is called a half turn, rotation of 90° is called a quarter turn.
- The collection of all the points in a plane, which are at a fixed distance from a fixed point in the plane, is called a circle. The fixed point is called the centre of the circle and the fixed distance is called the radius of the circle.
- If two points on a circle are joined by a line segment, then the line segment is called a chord of the circle.
- The chord, which passes through the centre of the circle, is called a diameter of the circle.
- Circles drawn in a plane with a common centre and different radii are called concentric circles.
- The area between the two concentric circles is known as circular ring
- Width of the circular ring(w) = $r_2 - r_1$

7th_Maths_Geometry_Chp4_New.indd 92 05-11-2019 17:36:21